

# L12. Adiabatic Quantum Computation

Xin Wan

Zhejiang University

# Outline

- Another Quantum Computing Paradigm
- The Landau-Zener Toy Model
- Standard Framework and Computational Power
- Bridging Gate and Adiabatic Models
- Core Application Areas

# Overview of Adiabatic Quantum Computing

- 1 **Encode the problem:** Encode the solution to a computational problem as the ground state of a physical system with Hamiltonian  $H_P$ .
- 2 **Prepare initial state:** Construct a simple initial Hamiltonian  $H_I$ , and prepare its ground state  $\psi_{gs}$ .
- 3 **Adiabatic evolution:** Let the system Hamiltonian change very slowly from  $H_I$  to  $H_P$ .
- 4 **Measure for solution:** After evolution, the system is in (or near) the ground state of  $H_P$ . Measuring it yields the problem's solution.

# Why Another Quantum Computing Paradigm?

- The quantum gate model
  - Like a digital circuit: Composed of sequences of discrete quantum logic gates (H, CNOT, Toffoli...).
  - Precisely control the evolution path of the quantum state.
- Challenges:
  - Sensitive to decoherence and operational errors.
  - Algorithm design for problems like combinatorial optimization and machine learning is not always intuitive.
- Alternative approach: Adiabatic Quantum Computing
  - An analog quantum computing paradigm.
  - Let an easily prepared quantum state *naturally* evolve into the solution of the target problem as the physical system's Hamiltonian changes slowly.

| Gate Model (Digital)                                             | Adiabatic Model (Analog)                                                         |
|------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Like building a complex structure step-by-step with Lego bricks. | Like slowly deforming a piece of clay from a simple shape into the target shape. |
| Precise control at each step.                                    | Control the global “deformation path.”                                           |
| Sensitive to errors in individual “steps.”                       | Potentially more robust to the “path” and “speed.”                               |

Physical Picture: Finding the “lowest energy valley” (optimal solution). Instead of jumping directly, we slowly morph the entire landscape (the system Hamiltonian) to guide the ball (system state) smoothly into the target valley.

# The Landau-Zener Toy Model

- We study adiabatic passage using the simplest quantum model — a qubit.
- The system is described by a Hamiltonian with a tunneling amplitude  $\hbar\Delta_{\min}$  controlled by an external bias field  $\hbar\varepsilon$ :

$$\mathcal{H}(\varepsilon) = -\frac{\hbar\varepsilon}{2}\sigma^z - \frac{\hbar\Delta_{\min}}{2}\sigma^x.$$

- Equivalent to a magnetic field  $\mathbf{B} = \Delta(\varepsilon)\mathbf{n}$  interacting with a quantum spin  $\mathbf{S} = \frac{1}{2}\hbar\boldsymbol{\sigma}$ .

# Ground State Representation

- We can compute the ground states for any value of  $\varepsilon$ :

$$\mathcal{H}(\varepsilon) = -\frac{1}{2}\hbar\Delta(\varepsilon)\mathbf{n}\cdot\boldsymbol{\sigma}, \text{ with } \begin{cases} \Delta = \sqrt{\varepsilon^2 + \Delta_{\min}^2} \geq |\Delta_{\min}|, \\ \mathbf{n} = \frac{1}{\Delta}(\Delta_{\min}, 0, \varepsilon). \end{cases}$$

- **Energy Gap:** The ground state energy is  $-\frac{1}{2}\hbar\Delta(\varepsilon)$ , separated from the excited state by a gap  $\hbar\Delta(\varepsilon)$ .
- **Ground State Vector:** A pure state with a Bloch vector aligned with the magnetic field,  $\langle\boldsymbol{\sigma}\rangle = \mathbf{n}$ .

# Wave Function Geometry

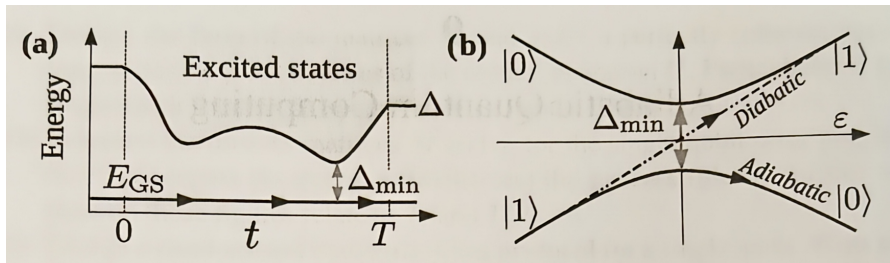
- The ground state wave function depends on the polar angle  $\mathbf{n} = (\sin \theta, 0, \cos \theta)$ :

$$|\tilde{0}_\varepsilon\rangle = \cos\left(\frac{\theta(\varepsilon)}{2}\right) |0\rangle + \sin\left(\frac{\theta(\varepsilon)}{2}\right) |1\rangle,$$

- $\varepsilon \ll -|\Delta_{\min}|$ : Ground state approximates  $|0, 0, -1\rangle \equiv |1\rangle$ .
- $\varepsilon \gg |\Delta_{\min}|$ : Ground state approximates  $|0, 0, +1\rangle \equiv |0\rangle$ .
- **Preparation:** These states can be prepared by waiting for relaxation or via measurement-and-reset protocols (e.g., microwave reset).

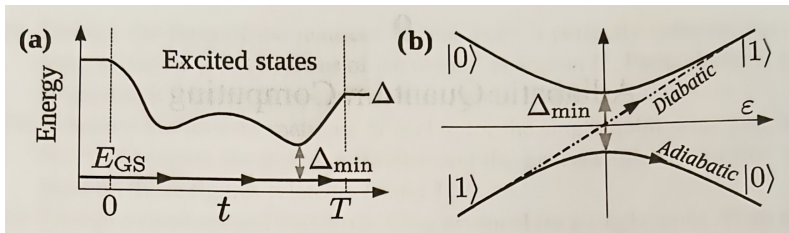


- We now change the control parameter  $\varepsilon$  as a function of time  $t$ . The critical question is: Does the qubit remain in the instantaneous ground state? Or does it get excited?



# Two Possible Scenarios

- 1 **Adiabatic Evolution:** The qubit's wave function is always the instantaneous ground state. There is no excitation.
- 2 **Diabatic Evolution:** The system fails to follow the instantaneous eigenstate. The qubit gets locked in the original state or becomes a superposition of both states.



# The Rotating Frame Hamiltonian

- To answer this, assume  $\varepsilon(t)$  grows monotonically and use a rotating frame that brings the instantaneous eigenstates to the computational basis:

$$|\psi(t)\rangle = e^{-i\theta(\varepsilon(t))\sigma^y/2}|\xi(t)\rangle.$$

- Using  $\varepsilon$  as a coordinate, the dynamics are governed by:<sup>1</sup>

$$\mathcal{H}_{\text{eff}} = - \left[ \frac{\hbar\Delta(\varepsilon)}{2}\sigma^z - \frac{\hbar\Delta_{\min}\dot{\varepsilon}}{2\Delta^2(\varepsilon)}\sigma^y \right].$$

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<sup>1</sup>See Appendix 12A

- Terminology
  - **Adiabatic term** ( $\propto \sigma^z$ ): Dominates in the perfectly adiabatic limit ( $\dot{\varepsilon} \rightarrow 0$ ).
  - **Counter-diabatic term** ( $\propto \sigma^y$ ): A weak perturbation that drives transitions between states.
- Perfect adiabatic limit,  $\dot{\varepsilon} \rightarrow 0$ :

$$|\xi(t)\rangle \rightarrow e^{i\phi(t)} |0\rangle, \quad \phi(t) = \int \Delta(\varepsilon) d\varepsilon / \dot{\varepsilon}.$$

- Corrections of order

$$\dot{\varepsilon} \Delta_{\min} / \Delta^2(\varepsilon) < \dot{\varepsilon} / \Delta_{\min}^2.$$

- When the speed of change  $\dot{\epsilon}$  is constant, the system admits an analytical solution for transition probability, known as the **Landau-Zener formula**,

$$P_{LZ} = \exp\left(-\frac{\pi}{2} \frac{\Delta_{\min}^2}{\dot{\epsilon}}\right) = \exp\left(-\frac{\pi}{2\omega_{evo}\tau_{adia}}\right).$$

- $\tau_{adia} = \Delta_{\min}^{-1}$ : Inverse minimum qubit frequency.
- $\omega_{evo} = \dot{\epsilon}/\Delta_{\min}$ : Hamiltonian's rate of change.
- Regimes
  - **Diabatic regime (fast):**  $P_{LZ} \rightarrow 1$  (Excitation is certain).
  - **Adiabatic regime (slow):**  $P_{LZ} \rightarrow 0$  (Excitation is negligible).

# General Framework

We need to concretely construct an evolution from a *simple problem* to a *complex problem*.

## 1 Initial Hamiltonian $H_I$ :

- Its ground state is a **known and easily prepared** quantum state.
- Typically non-commuting with  $H_P$ , e.g.,  $H_I = -\sum_i \sigma_i^x$ , whose ground state is the superposition of all qubits in  $|+\rangle$ .

## 2 Problem Hamiltonian $H_P$ :

- Its ground state encodes the answer to the computational problem we want to solve.
- E.g., for the Traveling Salesman Problem, its ground state corresponds to the shortest route.

## ① Constructing the time-dependent Hamiltonian

- The most common method is linear interpolation. Let  $s = t/T \in [0, 1]$ , then

$$H(s) = [1 - s]H_I + sH_P$$

- The system starts from the ground state of  $H_I$  and, as  $s$  goes from 0 to 1, adiabatically evolves to the ground state of  $H_P$ .

### Example: A Two-Level System

- $H_I = -\sigma^x$
- $H_P = -\sigma^z$
- $H(s) = -(1 - s)\sigma^x - s\sigma^z$

- Adiabatic QC consists of the following steps:
  - 1 **Encode:** Map the optimization/computational problem to be solved to a problem Hamiltonian  $H_P$ , whose ground state is the solution.
  - 2 **Initialize:** Prepare the ground state  $|\psi(0)\rangle$  of the initial Hamiltonian  $H_I$ .  $H_I$  does not commute with  $H_P$ , and its ground state is easy to prepare.
  - 3 **Evolve:** Let the system evolve slowly according to  $H(s) = (1 - s)H_I + sH_P$  over a total time  $T$  (during which  $s$  goes from 0 to 1).
  - 4 **Measure:** At time  $t = T$ , measure the final state  $|\psi(T)\rangle$ . According to the adiabatic theorem, with high probability it is the ground state of  $H_P$ , i.e., the solution.
  - 5 **Output:** Typically, export the measured classical bit string.



# Core Guarantee: The Adiabatic Theorem

- A time-dependent Hamiltonian  $H(t)$ .
- At any time  $t$ , it has a set of instantaneous eigenstates  $|\psi_n(t)\rangle$  and eigenvalues  $E_n(t)$  (energy levels). We focus on the **ground state**  $|\psi_0(t)\rangle$  (lowest energy).

## Theorem:

If the system starts in the instantaneous ground state  $|\psi_0(0)\rangle$ , and the Hamiltonian  $H(t)$  changes **sufficiently slowly**, then at any time  $t$  during the evolution, the system will remain in the instantaneous ground state  $|\psi_0(t)\rangle$  with very high probability.

# How Slow Is Sufficiently Slow?

- The system's evolution is governed by

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H(t) |\Psi(t)\rangle$$

Starting in the ground state  $|\psi_0(0)\rangle$ , a rapidly changing  $H(t)$  can drive the system to excited states  $|\psi_{n \neq 0}(t)\rangle$ .

- The probability of staying in the ground state depends on:
  - **Energy gap**  $\Delta(t) = E_1(t) - E_0(t)$ : The energy difference between the ground and first excited states. **A larger gap makes the system more stable**, less likely to transition.
  - **Coupling strength**  $|\langle \psi_1 | dH/dt | \psi_0 \rangle|$ : The strength with which the Hamiltonian change “mixes” different energy levels. More vigorous mixing makes transitions easier.

- **The Adiabatic Condition** (quantitative criterion):

$$\frac{|\langle \psi_1(t) | \frac{dH}{dt} | \psi_0(t) \rangle|}{[E_1(t) - E_0(t)]^2} \equiv \frac{\mathcal{C}(t)}{\Delta(t)^2} \ll 1$$

- $\Delta(t)$ : Instantaneous energy gap. The guardian of stability.
- $\mathcal{C}(t)$ : Instantaneous coupling strength. The source of perturbation.
- The condition that the rate of perturbation per unit energy spacing must be very small must hold for the entire evolution ( $0 \leq t = sT \leq T$ ).<sup>2</sup>

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<sup>2</sup>Note that  $dH/dt = (1/T)dH/ds = (1/T)(H_P - H_I)$ .

- Therefore, from the adiabatic condition, we can estimate that the total evolution time  $T$  must satisfy:

$$T \gg T_{\min} \propto \frac{1}{\Delta_{\min}^2}$$

where  $\Delta_{\min} = \min_t \Delta(t)$  is the **minimum energy gap** along the evolution path.

- Crucial conclusion: The time required for adiabatic evolution is primarily determined by the point in the process where the system's energy gap is narrowest. A smaller minimum gap requires a longer evolution time.

# Gap Closure and Quantum Phase Transitions

- In complex many-body systems, the energy gap  $\Delta(t)$  can vary during evolution, and may even become **very small** or approach zero at certain critical points.
- **Gap closure:** When  $\Delta_{\min} \rightarrow 0$ , the required time  $T \rightarrow \infty$ . This means adiabatic computation fails.
- Gap closure typically corresponds to the system undergoing a quantum phase transition (QPT), i.e., a phase transition driven by quantum fluctuations at zero temperature.
- Core challenge: Design evolution paths to avoid or handle situations where the gap closes (or is extremely small).

# Key Point: Finding a Suitable $H_P$

- Goal: Finding the bit string  $z$  that minimizes a cost function  $C(z)$  [or maximizes  $-C(z)$ ] is equivalent to finding the ground state of  $H_P$ .
- Core Mapping Relationship:
  - Encoding: Classical optimization problem  $\rightarrow$  Classical cost function:  $C(z)$ , where  $z$  is a bit string.
  - Quantization: Classical cost function  $C(z) \rightarrow$  Hamiltonian  $H_P$ , satisfying:  $\langle z | H_P | z \rangle = C(z)$ .

# Case Study: The Max-Cut Problem

- **Problem:** Given an undirected graph  $G = (V, E)$ , find a way to partition the vertex set  $V$  into two subsets that maximizes the number of cut edges.
- **Classical encoding:** Assign a binary variable  $z_i \in \{+1, -1\}$  to each vertex  $i$ , denoting its subset (e.g.,  $+1$  for group A,  $-1$  for group B).
- **Cost function:** For an edge  $(i, j)$ , it contributes  $+1$  if vertices are in different groups (i.e.,  $z_i \neq z_j$ ), otherwise 0. Goal: Maximize the cut size

$$C(z) = \sum_{(i,j) \in E} \frac{1}{2}(1 - z_i z_j)$$

- **Quantum encoding:** Each vertex  $i$  corresponds to a qubit. Its computational basis states  $|0\rangle$  and  $|1\rangle$  correspond to the classical values  $z_i = +1$  and  $z_i = -1$ , respectively.
- **Constructing  $H_P$ :** Note that  $\sigma_i^z |\psi_i\rangle = z_i |\psi_i\rangle$ . Replace the classical variables  $z_i$  in the cost function  $C(z)$  with the operator  $\sigma_i^z$ :

$$H_P = - \sum_{(i,j) \in E} \frac{1}{2} (1 - \sigma_i^z \sigma_j^z)$$

- For any computational basis state  $|z\rangle$ ,  $\langle z | H_P | z \rangle = -C(z)$ . Minimizing the expectation value of  $H_P$  is equivalent to maximizing the cut size  $C$ .



## • Choosing the Initial Hamiltonian $H_I$ :

- Must not commute with  $H_P$  to ensure a non-trivial evolution path.
- Its ground state should be easy to prepare.
- Standard choice: Transverse field  $H_I = -\sum_{i=1}^n \sigma_i^x$ .
- Its ground state is  $|+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}}(|0\rangle + |1\rangle)^{\otimes n}$ , a uniform superposition state, easy to prepare.

## • Complete Adiabatic Evolution Hamiltonian:

$$H(s) = (1-s)H_I + sH_P = -(1-s)\sum_i \sigma_i^x - s\sum_{(i,j)\in E} \frac{1}{2}(1 - \sigma_i^z \sigma_j^z)$$

where  $s = t/T$  varies from 0 to 1.

# The Classical Ising Model

- **Universal Paradigm:** Many combinatorial optimization problems (including traveling salesman) can be mapped to finding the ground state of a classical Ising model with Hamiltonian

$$H_{\text{Ising}} = \sum_i h_i \sigma_i^z + \sum_{i < j} J_{ij} \sigma_i^z \sigma_j^z$$

- $h_i$  is the local field,  $J_{ij}$  is the coupling strength.
  - $\sigma_i^z$  is the z-component of the  $i$ -th spin (taking values  $\pm 1$ ).
- **Quantization:** Treating  $\sigma_i^z$  as Pauli-Z operators turns this expression into the quantum mechanical operator  $H_P$ .

- In general, such a deceptively simple model belongs to a problem set known as spin glasses, found in the physics world of disordered magnetic systems. It has been shown that computing the ground state of this model is a family of NP-hard problems.
- **Physical Realization:** D-Wave devices essentially involve building a programmable quantum Ising model and encoding different problems by tuning parameters  $h_i$ ,  $J_{ij}$  through qubit-qubit dipolar interactions and external control fields.

# Equivalence in Computational Power

- Adiabatic quantum computing and the quantum gate model are polynomially equivalent.<sup>3</sup>
  - 1 **Adiabatic models can be efficiently simulated by gate models:** Any adiabatic evolution process can be approximately simulated using a polynomial number of quantum gates via methods like **Trotter-Suzuki decomposition**.
  - 2 **Gate models can be efficiently simulated by adiabatic models:** Any quantum circuit can be “compiled” into a sequence of local, adiabatically evolved Hamiltonians (**Feynman-Kitaev construction**), whose ground state encodes the circuit’s output.

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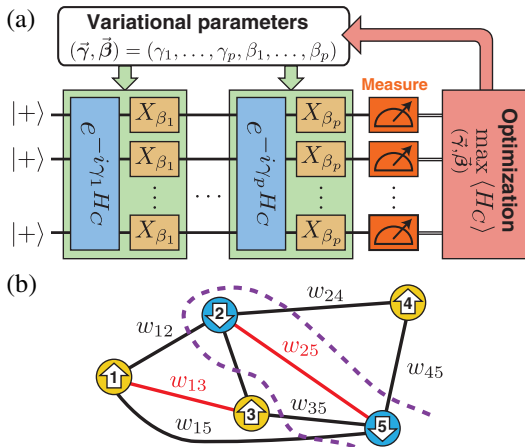
<sup>3</sup>Adiabatic quantum computing is another complete paradigm for universal quantum computation, theoretically possessing the same computational power (e.g., can solve all problems in BQP) as the gate model.

# Challenges in the NISQ Era

- Current devices have tens to hundreds of qubits but limited coherence times and noisy operations. Adiabatic computing faces a dilemma.
  - Precise adiabatic evolution requires long evolution time  
 $T \propto 1/\Delta_{\min}^2$ .
  - Long evolution fails due to decoherence and noise.
- What if we use shallow quantum circuits to approximately, not perfectly, pursue this evolution?

# Hybrid Quantum-Classical Architecture

- **Quantum processor:**  
prepare parameterized trial states.
- **Classical processor:**  
optimize the parameters to minimize energy.



# Quantum Approximate Optimization Algorithm

- Goal: Find a **parameterized quantum circuit** whose output state  $|\vec{\beta}, \vec{\gamma}\rangle$  is a **high-quality approximation** of the problem Hamiltonian  $H_C$ 's ground state.
- Inspiration: Adiabatic evolution is continuous, while quantum circuits are discrete. Use discrete gate sequences to mimic continuous adiabatic evolution.
- Essence of QAOA: A digital approximation of adiabatic quantum computing within the gate-model framework, serving as a key bridge connecting the two paradigms.

## • Quantum Circuit Construction

- Start from the initial state  $|\psi_0\rangle = |+\rangle^{\otimes n}$  (i.e., the ground state of  $H_{TF} = -\prod_i X_i$ ).
- Alternately apply unitary transformations generated by  $H_C$  and  $H_{TF}$ :

$$|\vec{\beta}, \vec{\gamma}\rangle = e^{-i\beta_p H_{TF}} e^{-i\gamma_p H_C} \dots e^{-i\beta_1 H_{TF}} e^{-i\gamma_1 H_C} |+\rangle^{\otimes n}$$

Here,  $p$  is the number of layers, and  $\vec{\beta} = (\beta_1, \dots, \beta_p)$  and  $\vec{\gamma} = (\gamma_1, \dots, \gamma_p)$  are tunable parameters.

- Measure this state multiple times, and compute expectation  $\langle H_C \rangle = \langle \vec{\beta}, \vec{\gamma} | H_C | \vec{\beta}, \vec{\gamma} \rangle$ .



- Loop until convergence:
  - Classical Optimizer (e.g., gradient descent): Input energy expectation value  $\langle H_C \rangle$ , adjust parameters  $(\vec{\beta}, \vec{\gamma})$  to minimize  $\langle H_C \rangle$ , and output new  $(\vec{\beta}, \vec{\gamma})$ .
  - Run quantum processor with new  $(\vec{\beta}, \vec{\gamma})$  and obtain new  $\langle H_C \rangle$  as feedback.
- Output: The state  $|\vec{\beta}^*, \vec{\gamma}^*\rangle$  corresponding to the optimized parameters  $(\vec{\beta}^*, \vec{\gamma}^*)$ . Measuring this state yields, with high probability, an approximate optimal solution for  $H_C$ .
  - When  $p \rightarrow \infty$ , QAOA approaches perfect adiabatic evolution.

| Feature          | AQC                                                 | QAOA                                                                            |
|------------------|-----------------------------------------------------|---------------------------------------------------------------------------------|
| Evolution        | Continuous, slow<br>Hamiltonian evolution<br>$H(s)$ | Discrete, alternating<br>gate sequence<br>$e^{-i\beta H_{TF}} e^{-i\gamma H_C}$ |
| Parameters       | Continuous schedule<br>function $s(t)$              | Discrete parameter set<br>$(\vec{\beta}, \vec{\gamma})$                         |
| Time/Depth       | Determined by min.<br>gap, can be long              | Fixed number of layers<br>$p$ , typically shallow                               |
| Output Fidelity  | Theoretically perfect<br>(if time sufficient)       | Approximate solution,<br>quality improves w/ $p$                                |
| NISQ Suitability | Poor (requires long<br>coherence)                   | Good (uses shallow<br>circuits)                                                 |

# Applications: Combinatorial Optimization

- **Natural Fit:** Many NP-hard combinatorial optimization problems can be naturally mapped to finding the lowest-energy spin configuration (ground state of an Ising model).
- **Current Status:** Specialized annealers (e.g., D-Wave) have shown potential on various problems, but a **universal quantum advantage** over the best classical algorithms for worst-case instances has not yet been definitively proven.

- **Explored Classical Problems:**

- Max-Cut, Vertex Cover
- Traveling Salesman Problem (TSP) and variants
- Boolean Satisfiability Problem (SAT)
- Integer Programming, Scheduling Problems

- **Industry-Relevant Applications:**

- Logistics and Route Planning: Package delivery, flight scheduling.
- Financial Modeling: Portfolio optimization, risk analysis.
- Chip Design: Circuit layout and routing.

# Applications: Quantum Simulation

- **Native Advantage:** Adiabatic evolution is itself a powerful tool for studying the ground states and dynamics of quantum many-body systems.
- **Significance:** Considered one of the most potentially disruptive application directions for adiabatic/quantum simulation.

- **Quantum Chemistry:**

- **Problem:** Solving the Schrödinger equation for molecular systems to obtain ground state energies and properties.
- **Method:** Use the molecular Hamiltonian directly as  $H_P$ , and prepare its ground state via adiabatic evolution.
- **Advantage:** May handle strongly correlated systems more efficiently than classical algorithms.

- **Condensed Matter Physics:**

- **Simulating novel states of matter:** e.g., high-temperature superconductors, quantum spin liquids, topological order.
- **Studying quantum phase transitions:** The adiabatic evolution path itself may cross a phase transition point, useful for studying critical behavior.

# Applications: Quantum Machine Learning

- **Core Idea:** Encode **optimization tasks** in machine learning as ground state search problems.
- **Potential:** Leveraging quantum parallelism and tunneling effects may accelerate time-consuming optimization steps in classical machine learning.

- **Specific Directions:**

- 1 **Training Quantum Neural Networks:** Transform the optimization of network weights into an adiabatic evolution to find optimal parameters.
- 2 **Quantum Boltzmann Machines:** The maximum likelihood estimation in their training can be transformed into a classical Ising model sampling problem, potentially accelerated by quantum annealing.
- 3 **Unsupervised Learning & Clustering:** e.g., mapping data point clustering to a Max-Cut problem.
- 4 **Support Vector Machines:** The training process involves a quadratic programming problem, mappable to an Ising model.



# Summary

*Adiabatic quantum computing is a robust and vibrant bridge connecting profound quantum physics principles with daunting real-world computational challenges.*

## 1 Physical Principle:

- Core is the Adiabatic Theorem.
- Key parameter is the minimum energy gap  $\Delta_{\min}$ , which determines computation time.
- Framework is the adiabatic evolution from the initial Hamiltonian  $H_I$  to the problem Hamiltonian  $H_P$ .

## 2 Computational Theory:

- Polynomially equivalent to the quantum gate model; a universal computing paradigm.
- Excels at combinatorial optimization and quantum simulation.
- QAOA is its bridging algorithm to the NISQ era.

## 3 Engineering Practice:

- Implemented via superconducting quantum annealing and Rydberg atom arrays.
- Faces three core challenges: gap closure, decoherence, and verification.
- Has broad application prospects in optimization, chemistry, materials science, and machine learning.

| Aspect                      | Gate Model                                          | Adiabatic Model                                                           |
|-----------------------------|-----------------------------------------------------|---------------------------------------------------------------------------|
| <b>Mindset</b>              | Digital/Logical                                     | Analog/Optimization                                                       |
| <b>Algorithm Design</b>     | Design clever gate sequences.                       | Design suitable Hamiltonians and evolution paths.                         |
| <b>Physical Realization</b> | Requires high-fidelity universal logic gates.       | Potentially more robust to specific interactions and decoherence.         |
| <b>Strengths</b>            | General algorithms (factoring, search, simulation). | Combinatorial optimization, ground state preparation, quantum simulation. |

# Appendix 12A: Solving Landau-Zener Model

Let us analyze the Landau-Zener Model

$$\mathcal{H}(\varepsilon) = -\frac{\hbar\varepsilon}{2}\sigma^z - \frac{\hbar\Delta_{\min}}{2}\sigma^x.$$

① Defining  $U = \exp(-i\theta(\varepsilon)\sigma^y/2)$ , show that

$$\mathcal{H}(\varepsilon) = -\frac{1}{2}\Delta(\varepsilon)U\sigma^zU^+.$$

- ② Rewrite the wave function of the two-level system in a rotating frame with  $|\psi(t)\rangle = U(t)|\xi(t)\rangle$  and show that the rotated state  $|\xi(t)\rangle$  evolves with the Hamiltonian:

$$\mathcal{H}_{\text{eff}} = - \left[ \frac{\hbar \Delta(\varepsilon)}{2} \sigma^z - \frac{\hbar \Delta_{\min} \dot{\varepsilon}}{2 \Delta^2(\varepsilon)} \sigma^y \right],$$

such that, with a change of variables  $t \rightarrow \varepsilon$  and  $\partial_t = \dot{\varepsilon} \partial_\varepsilon$ ,

$$i \partial_\varepsilon |\xi(\varepsilon)\rangle = - \left[ \frac{1}{2 \dot{\varepsilon}} \Delta(\varepsilon) \sigma^z - \frac{\Delta_{\min}}{2 \Delta(\varepsilon)^2} \sigma^y \right] |\xi(\varepsilon)\rangle.$$

- ④ Provide an upper bound on the prefactor in front of  $\sigma^y$ , compared with the  $\sigma^z$  term.
- ⑤ Approximately diagonalize the effective Hamiltonian, showing that the  $|0\rangle$  eigenstate of  $\sigma^z$  gets a correction of order  $\frac{\Delta_{\min}}{\Delta^2} \times \frac{\dot{\epsilon}}{\Delta} \leq \frac{\dot{\epsilon}}{\Delta_{\min}^2}$ .